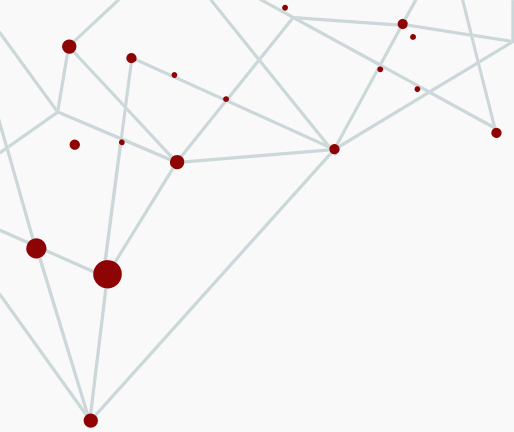
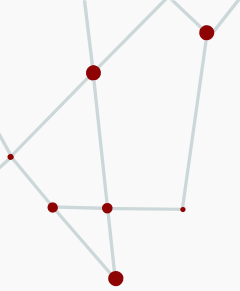




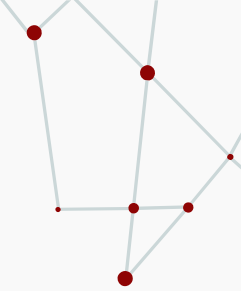
Self-Sovereign Identity (SSI)

By Musse Gebreezgabher





Introduction to SSI



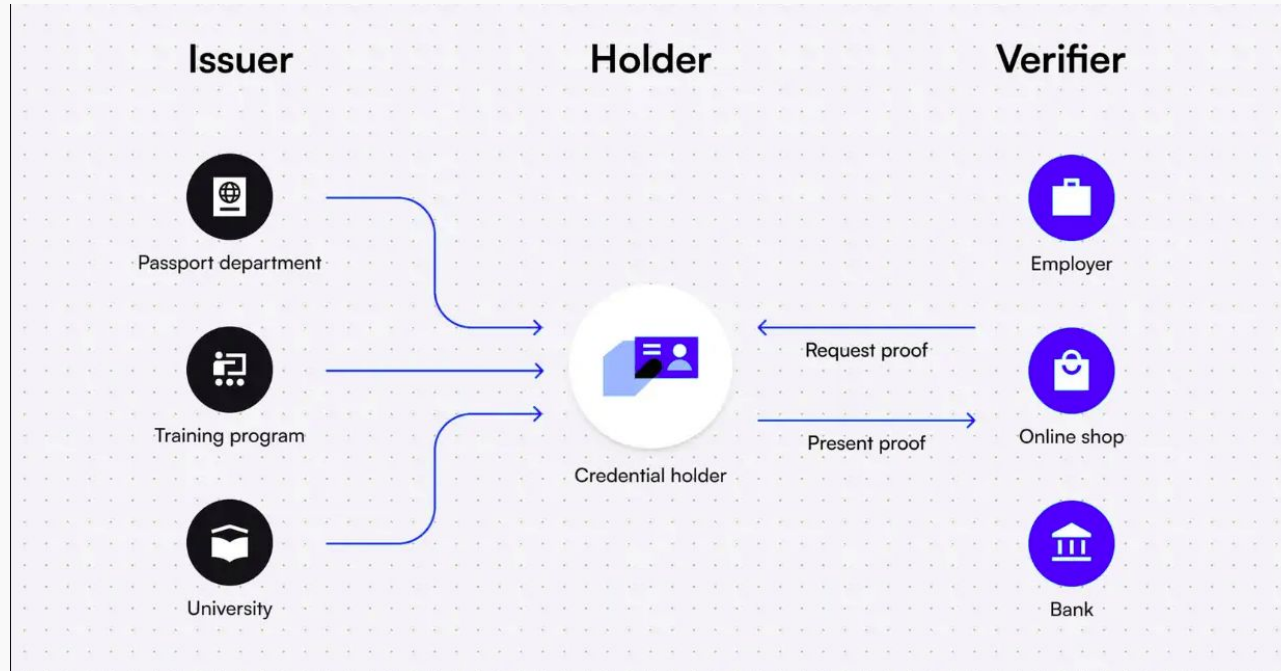
Self-Sovereign Identity (SSI) is a digital identity model that allows individuals to control and manage their personal data. It eliminates the need for centralized authorities like governments or corporations to store and verify identity information.



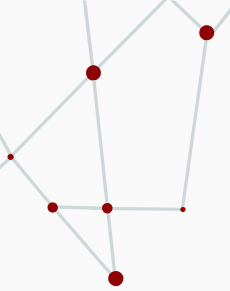
Participants in the SSI system

- **Holder:** Creates their decentralized identifier with a digital wallet and receives Verifiable Credentials.
- **Issuer:** Issues Verifiable Credentials. They sign it with their associated private key. VC also holds the DID.
- **Verifier:** Party checking the credential. Verifies whether the credential was signed by the issuer's DID.

Participants in the SSI system



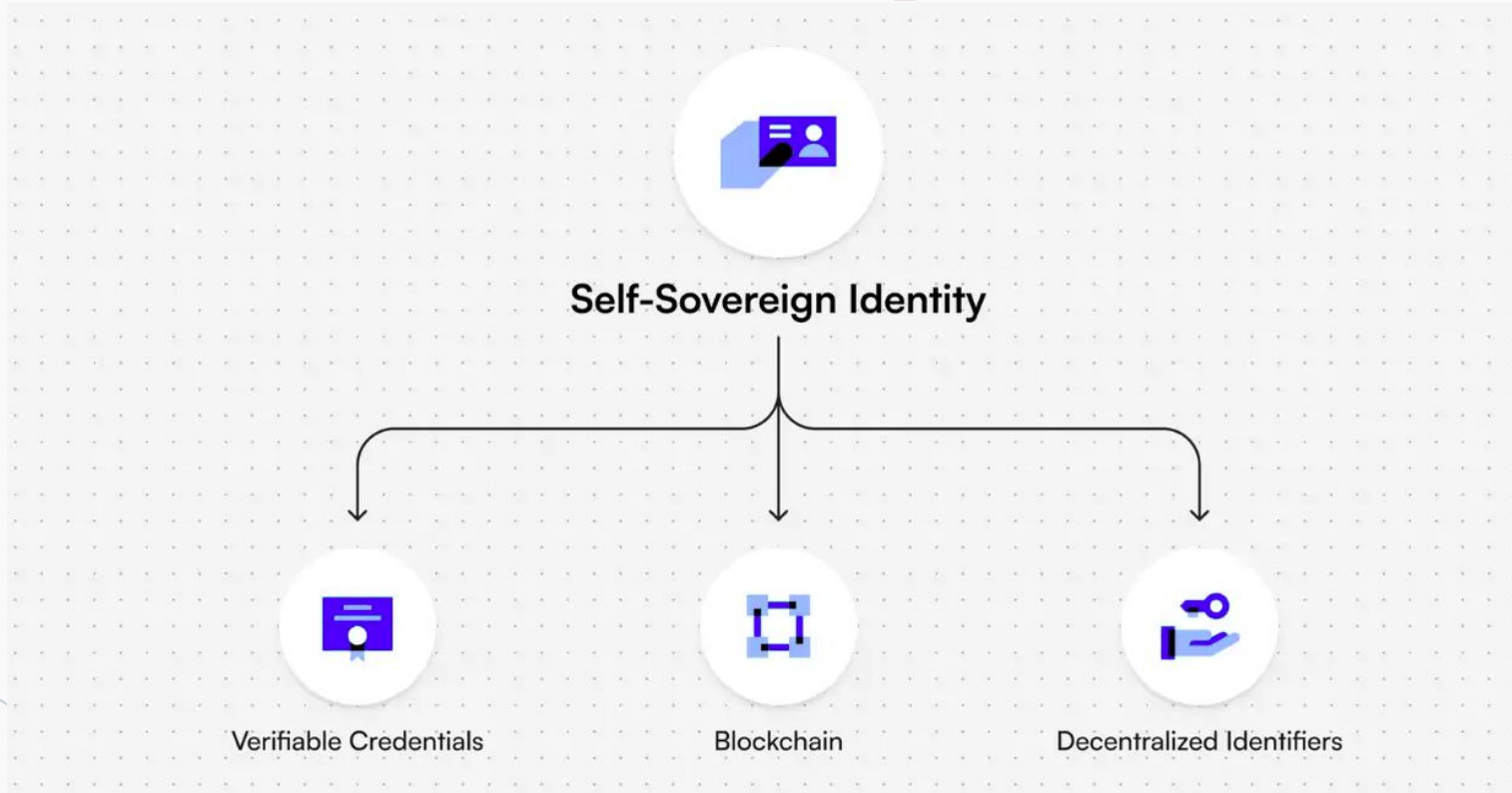
The trust triangle



3 Pillars of Self-Sovereign Identity

- **Blockchain:** Provides a tamper-proof ledger for identity transactions.
- **Decentralized Identifiers (DIDs):** Unique, globally resolvable identifiers stored on blockchain.
- **Verifiable Credentials (VCs):** Digitally signed credentials issued by trusted authorities.

3 Pillars of Self-Sovereign Identity



How SSI Works

- User creates a Decentralized Identifier (DID) using an SSI wallet.
- A trusted Issuer provides a Verifiable Credential (VC), signed with their private key.
- The Holder stores the VC securely in their digital wallet.
- When needed, the Holder presents a proof of the VC to a Verifier.
- The Verifier checks the validity of the VC by confirming the Issuer's DID and signature on the blockchain.

How Decentralized Identifiers (DIDs) Are Created

- The user generates a key pair (private and public key) using an SSI wallet.
- The public key is registered on a blockchain along with the DID Document.
- The DID Document contains cryptographic material for authentication.
- The blockchain ensures DID immutability and prevents tampering.
- A resolver fetches the DID Document when needed for verification.

How Verifiable Credentials (VCs) Are Issued

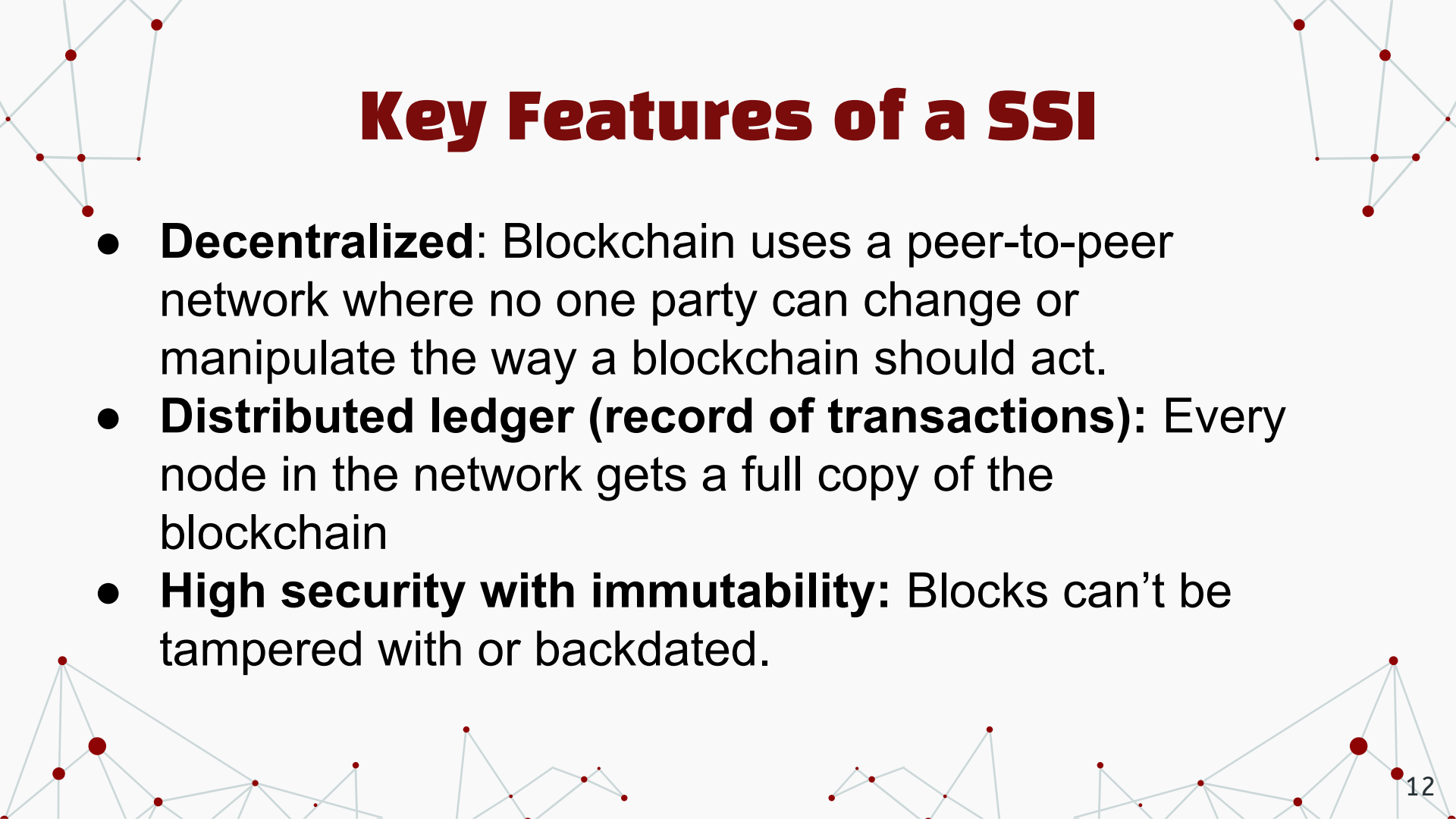
- The Holder requests a credential from a trusted Issuer (e.g., government, university).
- The Issuer verifies the Holder's identity using external authentication.
- The Issuer signs the Verifiable Credential with their private key and issues it.
- The Holder stores the VC in their digital wallet.
- The blockchain is used to verify the issuer's DID but does not store the actual credential.

How Credentials Are Verified

1. The Holder generates a Verifiable Presentation (VP) containing the credential.
2. The VP is cryptographically signed to prove authenticity.
3. The Verifier checks the signature against the Issuer's DID.
4. The Verifier uses a DID Resolver to fetch the Issuer's DID Document from the blockchain.
5. The verification process confirms that the credential is valid and unaltered.

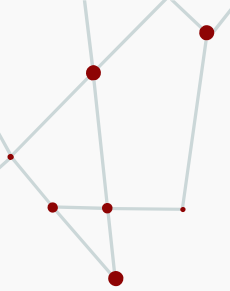
How Blockchain Supports SSI

- Stores Decentralized Identifiers (DIDs) in an immutable, verifiable way.
- Ensures integrity of DID Documents, preventing unauthorized changes.
- Does NOT store personal data but only cryptographic proofs.
- Provides a global resolution mechanism for retrieving identity data.



Key Features of a SSI

- **Decentralized:** Blockchain uses a peer-to-peer network where no one party can change or manipulate the way a blockchain should act.
- **Distributed ledger (record of transactions):** Every node in the network gets a full copy of the blockchain
- **High security with immutability:** Blocks can't be tampered with or backdated.



Advantages of SSI

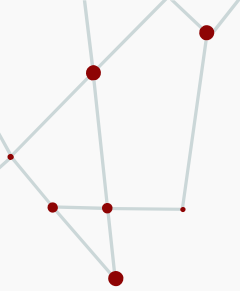
- User Control & Privacy: Users own their data without intermediaries.
- Reduced Reliance on Central Authorities: Eliminates single points of failure.
- Enhanced Security: Cryptographic signatures prevent forgery.
- Real-world Applications: Banking, healthcare, education, and travel.

Challenges and Criticisms of SSI

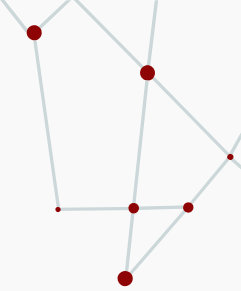
- Key Management: Users are responsible for protecting private keys, loss means identity loss.
- Legal Uncertainty: Regulations around self-sovereign identity are unclear.
- Dependence on Blockchain: Blockchain immutability is a double-edged sword.
- Adoption Hurdles: Governments and institutions may resist decentralization.

Comparison with Traditional Identity Models

- Centralized Identity: Managed by a single provider (e.g., government-issued IDs, social media logins).
- Federated Identity: Single Sign-On (SSO) using third-party authentication (e.g., Google, Facebook login).
- SSI: Users fully control their identity without intermediaries.
- Trade-offs: SSI offers security and privacy but at the cost of usability and key management.



Conclusion



- While SSI offers privacy and decentralization, usability and adoption remain challenges.
- Future research is needed to balance decentralization with practicality.

References

- Self-Sovereign Identity: The Ultimate Guide:
<https://www.dock.io/post/self-sovereign-identity>
- W3C Decentralized Identifiers (DIDs):
<https://www.w3.org/TR/did-core/>

The slide features decorative geometric patterns in the four corners. These patterns consist of light blue lines forming various polygons and triangles, with small dark red dots placed at the vertices. The central text 'Thank you!' is rendered in a large, bold, dark red serif font.

Thank you!